

SUIYANG LIAO

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Employment

Postdoctoral Research Fellow (The University of British Columbia)

01.2023-Present

Joint between Prof. Anna Blakney and Prof. Pieter Cullis

- Leading/supporting multi-lab collaborations to design LNPs for extrahepatic delivery of mRNA therapeutics
 - Prof. Colin Ross (Establishing luciferase reporter mouse model with single-point mutation)
 - Prof. Kenneth Harder (Preclinical assessment of the formulation)
 - Prof. Nicolas Bertrand (Isolation of protein corona via immunoprecipitation)
 - Prof. Leonard Foster (Proteomic workflow to analysis protein corona)
- Leading/supporting multi-lab collaborations to elucidate the pharmacokinetics of mRNA-LNP vaccines
 - Prof. Fabio Rossi (Muscle resident inflammatory cells and their role in LNP vaccines)
 - Prof. Kelly M. McNagny (Cellular response to mRNA-LNP formulations)
 - Prof. Leonard Foster (Mass spectrometry imaging for LNP tissue/cell tropism)

Interim Postdoc (École Polytechnique Fédérale de Lausanne)

04.2022-11.2022

Doctoral Assistant (École Polytechnique Fédérale de Lausanne)

09.2017-04.2022

Prof. Francesco Stellacci's laboratory

Education

École Polytechnique fédérale de Lausanne (EPFL)

Lausanne, Switzerland

Docteur en Sciences in Materials Science

09.2017-04.2022

Thesis: Single-chain nanoparticles for intracellular drug delivery

Advisor: Prof. Francesco Stellacci

Thesis Committee: Profs. Athina Anastasaki, Egbert Willem Meijer, and Darrell J. Irvine

Tsinghua University

Beijing, China

Bachelor of Engineering in Materials Science and Engineering

2011-2016

Bachelor of Business Administration

2013-2016

Core Experiences

Postdoctoral Researcher at UBC

Vancouver, Canada

Leading collaborative research to engineer lipid nanoparticles for extrahepatic mRNA delivery

Improving transfection potency of RNA therapeutics (mRNA, saRNA, circRNA) by investigating lipid-RNA interactions

Studying the relationships of LNP structure-protein corona-biodistribution

2023-Present

Supramolecular Nano-Materials and Interfaces Laboratory (EPFL)

Lausanne, Switzerland

Single-chain nanoparticles (SCNP) for intracellular delivery of biologics

2019-2022

Cellular uptake of SCNP topological isomers

2020-2021

Control and characterize the topology of SCNPs

2019-2021

Surface Characterization of bimetallic nanoparticles via MALDI-TOF

2018-2019

Protein corona on gold nanoparticles

2017-2018

ETH Zurich

Zürich, Switzerland

Guest Student in Analytical Chemistry (Advisor: Prof. Renato Zenobi)

2019

The University of Tokyo

Tokyo, Japan

Guest Student in Bioengineering (Advisor: Prof. Horacio Cabral)

2018

University of Oxford

Oxford, UK

Guest Student in Materials (Advisor: Prof. Andrew Watt)

2015

Fellowships, Awards, and Grants

2024/03	Nanomedicine Innovation Network (NMIN) Travel Award for CRS 2024 Annual Meeting (1,500 CAD)
2024/01	NMIN Travel Award for NMIN 2024 Research Conference (1,000 CAD)
2022	Co-applicant with Dr. Francesco Stellacci (EPFL) for an industrial funding to support a PhD position (~400,000 CHF)
2018	C2CPT International Internship Program, JSPS (~ 6,000 CHF to fund my summer intern at Utokyo)
2012-2016	Cyrus Tang scholarship, Cyrus Tang foundation (~15,000 RMB)
2011-2013	Tsinghua Xuetaang Talents Program (Top 3% of all Tsinghua students, ~10,000 RMB)

Invited/Selected talks

(Invited talk on behalf of Dr. Anna Blakney) Cystic Fibrosis Foundation Research Conference Title: Designing Lipid Nanoparticles for RNA therapeutics	Stevenson, WA, USA, Jun.2025
(Invited talk on behalf of Dr. Anna Blakney) TIDES USA: Oligonucleotide & Peptide Therapeutics. Title: Novel Design of Self-amplifying RNA and Formulation Challenges of Large RNA Payloads	San Diego, CA, USA, May.2025
(Invited webinar) Invited by Beckman Coulter Title: Transfection Potency of Lipid Nanoparticles Containing mRNA Depends on Relative Loading Levels	Online, Apr.2025
(Rising Star Lecture) CRSingapore2025 Title: Designing Lipid Nanoparticles for RNA therapeutics	Singapore, 2025, Feb.2025
(Trainee Seminar Award) Vancouver RNA club Title: Lipid nanoparticle design from mRNA vaccines to therapeutics	Vancouver, Canada, Sep.2024
(Invited talk) Presentation to Joerg Lahann lab at UMich	Online, Jul.2022
(Invited talk) Presentation to Jeremiah Johnson lab at MIT	Online, Jul.2022
(Selected talk) EPFL Initiative for Cancer Science and Engineering retreat	Lausanne, Switzerland, May.2022
(Invited talk) Presentation to Michael Mitchell lab at Upenn	Online, Apr.2022

Leadership

2025-Present	Leading trainee of the research teams of AVENGER aiming at developing next-generation m/saRNA-LNP vaccines
2023-2025	Leading trainee of the research teams funded by Nanomedicine Innovation Network for extrahepatic delivery of mRNA-LNPs (Roles: Communication across seven research labs with distinct backgrounds)
2016-2017	Head of class of ~30 students from the School of Materials, Tsinghua University (Roles: Organizing events and activities; Facilitating communication among classmates; Representing class in meetings)
2015-2016	Head of class of ~200 students at the School of Economy and Management, Tsinghua University (Roles: Took initiative in class projects; Fund raising; Organizing graduation ceremony)

Teaching

BMEG 400J/591J at School of Biomedical Engineering	UBC, Feb.2026
BMEG 580 (Gene therapy module) at School of Biomedical Engineering	UBC, Sep.2025
Guest Lecturer on the topic of nanoparticle characterization and medicinal usage (~45 first year Master's students)	
PHAR515 (Nanomedicines) at Faculty of Pharmaceutical Sciences	UBC, Feb.2025
Guest Lecturer on the topic of mRNA vaccine (~60 graduate students)	
ImmunoT MicroCourse (Vaccine and Infectious Diseases)	Online, Dec.2024
Guest Lecturer (~40 undergraduate students)	
MSE-245 (Soft Matter)	EPFL, 2021-2022
Teaching Assistant (~80 graduate students)	
Industry 4.0	Business School, Hohai University, 2017
Guest Lecturer (~100 students)	

Supervision/Mentoring (15 students of 7 nationalities)

2025/09-2026/04	Gargi Rao , Undergraduate Directed Studies, UBC Project: PEG-free LNP formulation and evaluation
2025/05-2025/09	Angeline Brasseur (co-supervised with Credo Casmil), Summer Intern from Polytech Nice Sophia Project: Novel saRNA constructs
2024/10-2025/04	Elli Chan and Sherinay Mahsut , SUS Mentorship Program, UBC
2023/08-2025/06	Shuangyu Wang , Research Assistant, UBC Projects: Fractionation of lipid nanoparticles Endosomal escape of lipid nanoparticles
2023/06-2024/03	Melody Cheng , Undergraduate Honor Thesis, UBC Project: Sequential assembly of lipid nanoparticles
2021/01-2021/07	Arthur Eliot Bouchez , Master Student, EPFL Project: Ligand functionalized single chain nanoparticles for intracellular drug delivery
2021/04-2021/08	Tina Torabi , Summer Intern, EPFL
2020/06-2021/02	Laura Cabizzosu , Master Student, EPFL Project: Drug Loading into single chain nanoparticles
2019/01-2019/06	Jiaying Hou , Master Student, EPFL Project: Surface ratio of alloy nanoparticles characterized via MALDI-TOF
2018/12-2019/05	Edgar Smit , Bachelor Student, EPFL Project: Surface composition evolution of gold and silver alloy nanoparticles
2018/07-2019/01	Océane Fort , Master Student, EPFL Project: Single-chain polymer nanoparticle synthesis
2018/01-2018/06	Jingwen Wang (co-supervised with Dr. Zhi Luo), Master Student, EPFL Project: Temperature and pH responsive single chain polymeric nanoparticle
2016/05-2017/05	Yusen Zhao and Jie Ren , Tsinghua University Project: Draw-spinning of polymer fibers

Presentations (6 invited talks; 6 selected talks; 11 posters)

1. (People's Choice Poster Award Top 3/800) CRS 2026 Annual Meeting & Exposition	<i>Lisbon, Portugal, Jul.2026</i>
2. (Oral presentation) Vancouver Area Mass Spectrometry	<i>Vancouver, Canada, Nov.2025</i>
3. (Oral presentation) MSL Seminar Series	<i>Vancouver, Canada, Sep.2025</i>
4. (Poster) Canadian Society of Pharmacology and Therapeutics Annual Meeting	<i>Vancouver, Canada, Jun.2025</i>
5. (Invited talk) Cystic Fibrosis Foundation Research Conference	<i>Stevenson, WA, USA, Jun.2025</i>
6. (Oral presentation) Diabetes & Obesity Research Day 2025	<i>Kelowna, Canada, May.2025</i>
7. (Invited talk) TIDES USA: Oligonucleotide & Peptide Therapeutics	<i>San Diego, CA, USA, May.2025</i>
8. (Invited webinar) Invited by Beckman Coulter	<i>Online, Apr.2025</i>
9. (Rising Star Lecture) CRSingapore2025	<i>Singapore, Feb.2025</i>
10. (Trainee Seminar Award) Vancouver RNA club	<i>Vancouver, Canada, Sep.2024</i>
11. (Top Poster Award) CRS 2024 Annual Meeting & Exposition	<i>Bologna, Italy, Jul.2024</i>
12. (Participation) XRay Diffraction & Scattering Summer School in the Canadian Light Source	<i>Saskatoon, Canada, Jun.2024</i>
13. (Poster) Nanomedicine Innovation Network 2024 Research Conference	<i>Vancouver, Canada, Jan.2024</i>
14. (Poster Judging) Vancouver Nanomedicine Day 2023	<i>Vancouver, Canada, Nov.2023</i>
15. (Poster) International School of Materials for Sustainable Development & Energy	<i>Erice, Italy, Jul.2022</i>
16. (Poster) Gordon Research Conference Bioinspired Materials	<i>Les Diablerets, Switzerland, Jun.2022</i>
17. (Invited talk) Presentation to Joerg Lahann lab at UMich	<i>Online, Jul.2022</i>
18. (Invited talk) Presentation to Jeremiah Johnson lab at MIT	<i>Online, Jul.2022</i>
19. (Selected talk) EPFL Initiative for Cancer Science and Engineering retreat	<i>Lausanne, Switzerland, May.2022</i>
20. (Invited talk) Presentation to Michael Mitchell lab at UPenn	<i>Online, Apr.2022</i>
21. (Poster) EDMX research day at EPFL	<i>Online, Feb.2021</i>
22. (Pitch talk and poster) EDMX research day at EPFL	<i>Lausanne, Switzerland, Nov.2019</i>
23. (Poster) Bioinspired Materials: from understanding, through processing, to replication	<i>Monte Verità, Switzerland, Oct.2019</i>
24. (Poster) FEBS workshop: Hydrodynamic and thermodynamic analysis	<i>Prague, Czech Republic, Sep.2018</i>
25. (Poster) CCMX workshop Nanoparticles: from Fundamentals to applications in Life Science	<i>Kandersteg, Switzerland, Jan.2018</i>

Outreach/Service

(Popular science article) Liao, S. "How nanomedicine gets inside your cells and treats you from the inside out."
<https://doi.org/10.64628/AAM.yxntcp3yp>. *The Conversation Canada*, Apr. 2026

(Poster Judge) Control Release Society 2024 Annual Meeting and Expo *Lisbon, Portugal, Jul.2026*

(Event Facilitator) UBC LSI Postdoc Breakfast Social *Vancouver, Canada, Mar.2025*

(Workshop Facilitator) UBC Postdoc Industry and Entrepreneurship workshops *Vancouver, Canada, Nov.2024-Jan.2025*

(Science Slam Presenter) Science Slam Canada: Slam into Fall *Vancouver, Canada, Sep.2024*

Representative for postdoctoral fellows at LSI Executive Committee *Vancouver, Canada, May.2025-Present*

UBC Michael Smith Laboratories Postdoc Committee *Vancouver, Canada, May.2025-Present*

(Co-reviewer of journal articles) Nature Communication; Journal of Controlled Release; Small

(Grant evaluation) FONDECYT (Chile) *Aug.2025*

Professional Societies

Controlled Release Society *2024-Present*

RNA Society *2025-Present*

Selected Publications

Full list via [Google Scholar](#). (h-index=15 with seven publications as first and/or corresponding author)

- (1)
Liao, S.; Fox, K.; Birkenshaw, A.; Wadhwa, A.; Wang, S.; Badard, K.; Gaudreault, N.; Rekve, J.; Cheng, M. H. Y.; Bertrand, N.; Foster, L. J.; Harder, K. W.; Ross, C. J. D.; Badea, I.; Blakney, A. K.; Cullis, P. R. Liposomal Lipid Nanoparticles Containing Dihydrosphingomyelin Exhibit Improved Stability and Extended Hepatic and Extrahepatic Transfection. *Journal of Controlled Release* **2026**, 396, 115098. <https://doi.org/10.1016/j.jconrel.2026.115098>.
- (2)
Casmil, I. C.; Jin, J.; Won, E.-J.; Huang, C.; Liao, S.; Cha-Molstad, H.; Blakney, A. K. The Advent of Clinical Self-Amplifying RNA Vaccines. *Molecular Therapy* **2025**, 0 (0). <https://doi.org/10.1016/j.ymthe.2025.03.060>.
- (3)
Liao, S.; Wang, S.; Wadhwa, A.; Birkenshaw, A.; Fox, K.; Cheng, M. H. Y.; Casmil, I. C.; Magana, A. A.; Bathula, N. V.; Ho, C. H.; Cheng, J.-Y.; Foster, L. J.; Harder, K. W.; Ross, C. J. D.; Cullis, P. R.; Blakney, A. K. Transfection Potency of Lipid Nanoparticles Containing mRNA Depends on Relative Loading Levels. *ACS Appl. Mater. Interfaces* **2024**. <https://doi.org/10.1021/acsami.4c20077>.
- (4)
Chen, S. P.; Wang, S.; Liao, S.; Blakney, A. K. Exploring the Effects of Incorporating Different Bioactive Phospholipids into Messenger Ribonucleic Acid Lipid Nanoparticle (mRNA LNP) Formulations. *ACS Bio Med Chem Au* **2024**. <https://doi.org/10.1021/acsbioimedchemau.4c00085>.
- (5)
Bathula, N. V.; Friesen, J. J.; Casmil, I. C.; Wayne, C. J.; Liao, S.; Soriano, S. K. V.; Ho, C. H.; Strumpel, A.; Blakney, A. K. Delivery Vehicle and Route of Administration Influences Self-Amplifying RNA Biodistribution, Expression Kinetics, and Reactogenicity. *Journal of Controlled Release* **2024**, 374, 28–38. <https://doi.org/10.1016/j.jconrel.2024.07.078>.
- (6)
G. Popova, P.; Chen, S. P.; Liao, S.; Sadarangani, M.; Blakney, A. K. Clinical Perspective on Topical Vaccination Strategies. *Advanced Drug Delivery Reviews* **2024**, 208, 115292. <https://doi.org/10.1016/j.addr.2024.115292>.
- (7)
Liao, S.; Wei, L.; Bouchez, A. E.; Stellacci, F. Influence of Structural Dynamics on Cell Uptake Investigated with Single-Chain Polymeric Nanoparticles. *Chem* **2023**, 9 (6), 1562–1577. <https://doi.org/10.1016/j.chempr.2023.03.012>.
- (8)
Liao, S.; Luo, Z.; Metternich, J. B.; Zenobi, R.; Stellacci, F. Quantification of Surface Composition and Segregation on AuAg Bimetallic Nanoparticles by MALDI MS. *Nanoscale* **2020**, 12 (44), 22639–22644. <https://doi.org/10.1039/D0NR05061J>.
- (9)
Tao, A.; Huang, G. L.; Igarashi, K.; Hong, T.; Liao, S.; Stellacci, F.; Matsumoto, Y.; Yamasoba, T.; Kataoka, K.; Cabral, H. Polymeric Micelles Loading Proteins through Concurrent Ion Complexation and pH-Cleavable Covalent Bonding for In Vivo Delivery. *Macromolecular Bioscience* **2020**, 20 (1), 1900161. <https://doi.org/10.1002/mabi.201900161>.
- (10)
Bekdemir, A.; Liao, S.; Stellacci, F. On the Effect of Ligand Shell Heterogeneity on Nanoparticle/Protein Binding Thermodynamics. *Colloids and Surfaces B: Biointerfaces* **2019**, 174, 367–373. <https://doi.org/10.1016/j.colsurfb.2018.11.027>.
- (11)
Bai, X.; Liao, S.; Huang, Y.; Song, J.; Liu, Z.; Fang, M.; Xu, C.; Cui, Y.; Wu, H. Continuous Draw Spinning of Extra-Long Silver Submicron Fibers with Micrometer Patterning Capability. *Nano Lett.* **2017**, 17 (3), 1883–1891. <https://doi.org/10.1021/acs.nanolett.6b05205>.
- (12)
Liao, S.; Bai, X.; Song, J.; Zhang, Q.; Ren, J.; Zhao, Y.; Wu, H. Draw-Spinning of Kilometer-Long and Highly Stretchable Polymer Submicrometer Fibers. *Advanced Science* **2017**, 4 (9), 1600480. <https://doi.org/10.1002/advs.201600480>.

References

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